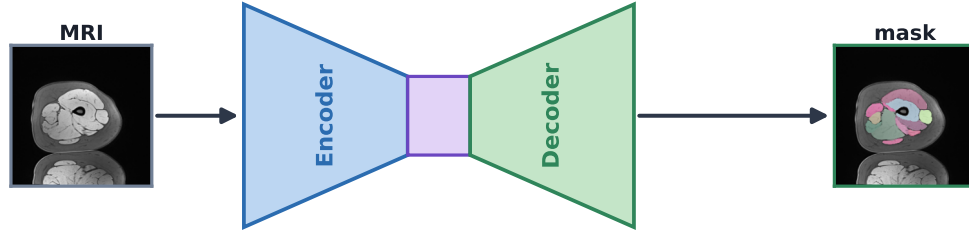
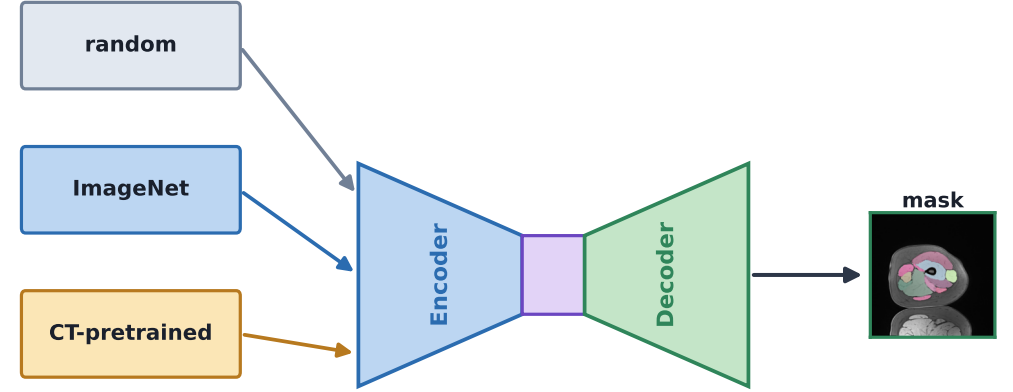


a Factor 1 — segmentation architecture



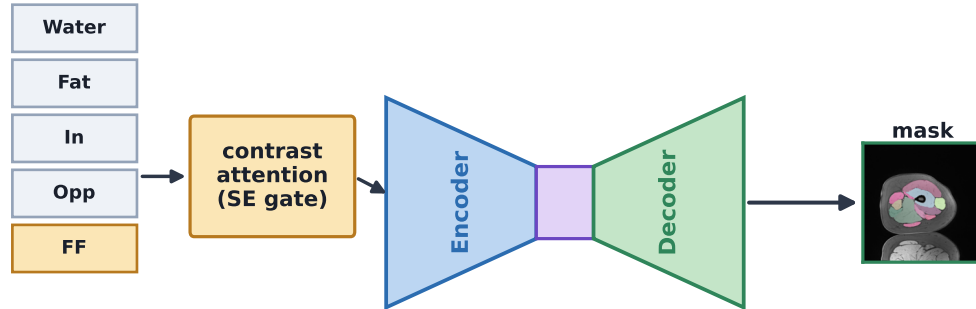
$backbone \in \{ U-Net, U-Net++, DeepLabV3+, SegFormer \}$

b Factor 2 — encoder pretraining source



only the encoder initialisation changes

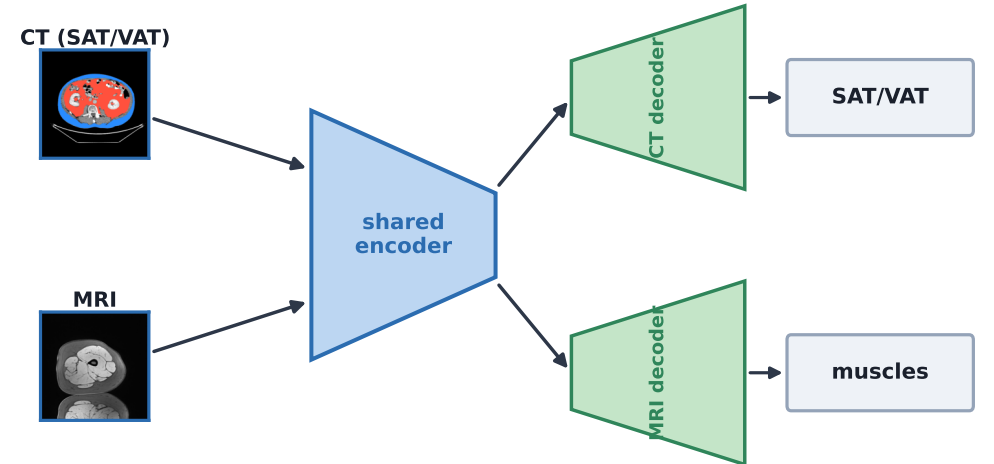
c Factor 3 — physics-guided input design



$$s = \alpha W_2 \delta(W_1 \text{GAP}(x)), \quad \hat{x} = s \odot x$$

per-image gate $s \in [0,1]$ per contrast reweights each channel; $FF = \text{Fat}/(\text{Fat}+\text{Water})$

d Factor 4 — multi-task (shared encoder)



tied encoder weights; robust to normalisation scheme